

Multiple Choice (10 marks)

1. What is the rank of the following matrix? $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
2. __ refers to a model that can neither model the training data nor generalize to new data.
 - (a) good fitting
 - (b) overfitting
 - (c) underfitting
 - (d) all of the above
3. Compared to the variance of the Maximum Likelihood Estimate (MLE), the variance of the Maximum A Posteriori (MAP) estimate is _____
 - (a) higher
 - (b) same
 - (c) lower
 - (d) it could be any of the above
4. Existential risks posed by AI are most commonly associated with which of the following professors?
 - (a) Nando de Freitas
 - (b) Yann LeCun
 - (c) Stuart Russell
 - (d) Jitendra Malik
5. The numerical output of a sigmoid node in a neural network:
 - (a) Is unbounded, encompassing all real numbers.
 - (b) Is unbounded, encompassing all integers.
 - (c) Is bounded between 0 and 1.
 - (d) Is bounded between -1 and 1.

True / False (10 marks)

Answer True or False

6. True or False: To calculate $P(H|E, F)$, the sufficient numbers include $P(E, F)$, $P(H)$, and $P(E, F|H)$.
7. True or False: A Bayesian network structured as $H \rightarrow U \leftarrow P \leftarrow W$ requires only 4 independent parameters to fully define its joint distribution.
8. True or False: Industrial-scale neural networks are typically trained on GPUs, which are optimized for parallel processing tasks.
9. True or False: Both Ridge and Lasso regression are equally effective for feature selection in all scenarios.
10. True or False: The model obtained by applying linear regression may differ when using forward stage wise selection compared to the end of subset identification.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether all the bits are unset in the given range or not.
12. Write a function to check if the given expression is balanced or not.

End of Paper